



AER1416: Numerical Methods for Uncertainty Quantification

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List of Case Studies

P1 Consider the quadratic function in d variables defined below:

$$f(x) = a_0 + \sum_{i=1}^d a_i x_i + \gamma \sum_{i=1}^d \sum_{j=1}^d a_{ij} x_i x_j,$$

where $x_i, i = 1, 2, \dots, d$ are independent random variables drawn from $\mathcal{N}(0, 1)$, γ is a scaling factor, $a_0 = 1$, $a_i = \frac{1}{\sqrt{i}}$, and $a_{ij} = \begin{cases} \frac{1}{i} & \text{if } i = j \\ \frac{\min(i,j)}{\max(i,j)^2(i+j)} & \text{if } i \neq j. \end{cases}$

P2 Consider the following second-order ordinary differential equation

$$m\ddot{u} + c\dot{u} + ku = f(t),$$

subject to the initial conditions $u(0) = \dot{u}(0) = 0$. $m \sim \mathcal{U}[0.9, 1.1]$, $k \sim \mathcal{U}[10, 12]$ and $c \sim \mathcal{U}[0.8, 1.2]$ are independent random variables. The forcing term is defined as $f(t) = F_0 \cos(\omega t)$, where $\omega \sim \mathcal{U}[0.1, 0.2]$ and $F_0 \sim \mathcal{U}[0.9, 1.1]$. All the random variables within this model are assumed to be statistically independent.

You can use any time-marching scheme of your choice to construct the forward model.

P3 Consider the linear *random* algebraic system of equations arising from spatial discretization of the steady-state two-dimensional stochastic diffusion equation

$$\mathbf{A}(\boldsymbol{\theta})\mathbf{u}(\boldsymbol{\theta}) = \mathbf{b},$$

where $\mathbf{A}(\boldsymbol{\theta}) = \mathbf{A}_0 + \sum_{i=1}^d \theta_i \mathbf{A}_i$, with $\mathbf{A}_i \in \mathbb{R}^{n \times n}$, $i = 0, 1, 2, \dots, d$ denoting deterministic matrices, $\theta_i \sim \mathcal{U}[-3, +3]$, $i = 1, 2, \dots, d$ are independent and identically distributed random variables, $\mathbf{b} \in \mathbb{R}^n$ is a deterministic vector, and $\mathbf{u}(\boldsymbol{\theta}) \in \mathbb{R}^n$ denotes the discretized solution vector.

- A MATLAB data file **P3.mat** can be downloaded from this folder that contains the matrices $\mathbf{A}_i, i = 0, 1, 2, 3$ and the right hand side vector $\mathbf{b}$ for the case when $d = 3, n = 143$.
- Two MATLAB functions **PlotP3_3D.m** and **PlotP3_contour.m** are provided to plot the discrete solution $\mathbf{u}$ on the spatial mesh. These functions can be used to visualize the nominal solution, mean, and variance of $\mathbf{u}$.